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# Vyshak Athreya Bellur Keshavamurthy

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## OBJECTIVE

Seeking challenging opportunity to scale skills and generate value for the company

## EDUCATION

**San Diego State University**,
San Diego, CA
Master of Science in Computer Science
August 2016 - May 2018

**Visvesvaraya Technological University**,
Karnataka, India
Bachelor in Electronics & Communication Engineering
July 2010 - June 2014

## GRADUATE COURSEWORK

- iOS app development
- Android app development
- Computer & Data networks
- Advanced Web Development
- Machine Learning
- Data Mining
- Speech Processing
- Spatial Databases

## COURSERA CERTIFICATIONS

- Algorithms Specialization
- Python data structures

## TECHNICAL SKILLS

**Languages:** Java, Python, Swift, C, JavaScript, SQL, HTML5, CSS3, XML

**Frameworks:** Spring, EJB, UIKit, Map Kit, Core Location, Alamo fire, Google Maps, Volley, Bootstrap, Angular JS

**Libraries:** Keras, Tensor flow, SciPy, DTW, NumPy, Matplotlib, Librosa

**Algorithms:** SVM, Naive Bayes, KNN, K-Means, PCA, Regression

**Tools**: Eclipse, MATLAB, Android Studio, XCode, GIT, SVN, MySQL Workbench, Argo UML, Anaconda

## WORK EXPERIENCE

*HCL Technologies, Karnataka, India | Sep 2014 - Aug 2016*

Software Engineer

- Debugged, gave enhancements to a trading application built on MVC framework in a team of 10 using **vectors, hash map, swing API and XML**
- Learned about **Agile** model following **scrum** methodology, version control using **SVN** and issue workflow in **JIRA**
- Created Business Requirement Document using ArgoUML in a team of 14
- Facilitated automation by developing a web portal using **Spring, MySQL, JSP, servlet, JDBC, HTML5, CSS3, and JavaScript**
- Reduced process delays by 60% (5 hours per day) by implementing excel creation & parsing using Apache POI
- Rewarded with **HCL Spot Award** for successfully completing phase I of Project Management Office Automation

## ACADEMIC PROJECTS

- In IOS and Android platforms **(Java and Swift)** created,
  - A finance app to monitor expenses in multiple currencies and alert different levels of budget consumption using firebase, shared preferences, multithreading, JSON, Volley and Alamo fire.
  - A chat application with the location-based search option. Used firebase, Google maps API, MapKit, CoreLocation. Fixed errors using USB debugging, debug gauges, breakpoints and log messages.
- Designed a drawing application for Android OS **(Java)**. Performed motion detection and computed swipe velocity.
- Implemented a card matching game for iOS **(Swift)** based on MVC design pattern, following Human Interface Guidelines. Made use of optionals, property observers, structs, classes, tuples, and dictionary. Adapted to different devices with stack view embedding and auto layout.
- Recognized simple speech commands **(Python)** with 70% efficiency using K-means, Mel-power spectrograms, Gaussian Mixture Model, dynamic time warping, Keras, LSTM, L1, L2 and dropout techniques. A team of 5 members.
- Programmed scripts (**Python)** to simulate memory management and scheduling. Achieved best memory management using Least Recently Used algorithm. With Frequency Scaling, improved scheduling by 30%.
- Developing kernels **(Python)** based on PCA, K-means and SVM models to find the percentage of similarity and origin of language scripts.
- Built a data-logging desktop application **(Python)** using NumPy, SciPy, lab jack device drivers, Tkinter, matplotlib, and xlrd package
- Explored the impact of L1, L2, and dropout regularizers on feed-forward neural networks **(Python)**. Implemented automatic caching of the PCA analysis and feature generation and simplified Cross validation process.
- Experimented with Recurrent Neural Networks architectures **(Python)** such as LSTM, GRU and simple RNN. Evaluated results using Tensor board scalars. Reduced errors by !4% by using Voice Activity Detector.
- Created web applications ( http://cmems2018.com , http://grapheton.com and http://jadran.sdsu.edu/~jadrn058/proj1 )using **Perl, HTML 5, CSS 3, JavaScript, AJAX, and MySQL.** Implemented session management.